Knight Moves Grid

Time limit: 1.00 s
Memory limit: 512 MB

CSES 3217 | Introductory Problems | https://cses.fi/problemset/task/3217

There is a knight on an $n \times n$ chessboard. For each square, print the minimum number of moves the knight needs to do to reach the top-left corner.

Input

The only line has an integer n.

Output

Print the number of moves for each square.

Constraints

- $4 \le n \le 1000$

Example

Input
8

Output
0 3 2 3 2 3 4 5 3 4 1 2 3 4 3 4 2 1 4 3 2 3 4 5 3 2 3 2 3 4 3 4 2 3 2 3 4 3 4 5 3 4 3 4 3 4 5 4 4 3 4 3 4 5 4 5 5 4 5 4 5 4 5 6